

Chapter 3

Variate Transformations

3.1 Variate Transformation

Let T be any statistic which is asymptotically normally distributed with mean θ and variance $\Psi(\theta)$, where $\Psi(\theta)$ is some function of the parameter θ , i.e., $T \sim N\{\theta, \Psi(\theta)\}$. Let us transform T by a function g as $g(T)$ where g is a function which possesses first order derivative which is continuous and $g'(\theta) \neq 0$, where $(')$ denotes differentiation with respect to the parameter θ . Then $g(T)$ is normally distributed about mean $g(\theta)$ and variance $[g'(\theta)]^2 \cdot \Psi(\theta)$, i.e.,

$$g(T) = N\left[g(\theta), [g'(\theta)]^2 \cdot \Psi(\theta)\right], \quad (3.1)$$

asymptotically, provided $g'(\theta) \neq 0$ is continuous in the neighbourhood of θ .

In general $Var(T) = \Psi(\theta)$ will be dependent on the parameter θ . We are interested in obtaining a function g such that the asymptotic variance of the transformed statistic $g(T)$ is independent of θ , i.e., it should be constant. In other words, we want g such that $Var[g(T) = [g'(\theta)]^2] \cdot \Psi(\theta) = \text{Constant} = c^2$, (say) i.e.,

$$\left|g'(\theta)\right| = \frac{c}{\sqrt{\Psi(\theta)}},$$

Integrating both sides with respect to θ , we get

$$g(\theta) = \int \frac{c}{\sqrt{\Psi(\theta)}} d\theta. \quad (3.2)$$

3.2 Types of Variate Transformation

The commonly used transformations are:

- (i) Square root transformation,
- (ii) Sine inverse ($\sin^{-1}$) transformation,
- (iii) Logarithmic transformation, and
- (iv) Fisher's Z -Transformation.

3.2.1 Square Root Transformation

Square root transformation is a transformation for the Poisson variate. If a variable X follows Poisson distribution with parameter λ (assumed to be large), then we know that asymptotic distribution of X is normal (as $\lambda \rightarrow \infty$) with $E(X) = \lambda$ and $Var(X) = \lambda = \Psi(\lambda)$, in the above notations. Then (3.2) gives the function

$$g(\lambda) = \int \frac{c}{\sqrt{\lambda}} d\lambda = 2c\sqrt{\lambda}.$$

Now we select c in such a way that $2c = 1$, i.e., $c = \frac{1}{2}$ so that $g(\lambda) = \sqrt{\lambda}$.

Hence the transformed variable is $g(X) = \sqrt{X}$. Using (3.1), the transformed variable $\sqrt{X}$ has the mean

$$g(\lambda) = \sqrt{\lambda}$$

and variance

$$Var(\sqrt{X}) = [g'(\lambda)]^2 \cdot \Psi(\lambda)$$

$$\begin{aligned}
&= \left[\frac{1}{2\sqrt{\lambda}} \right]^2 \cdot \lambda \\
&= \frac{1}{4}.
\end{aligned}$$

Or alternatively

$$Var\left(\sqrt{X}\right) = c^2 = \left(\frac{1}{2}\right)^2 = \frac{1}{4}.$$

Hence $\sqrt{X} \sim N\left(\sqrt{\lambda}, \frac{1}{4}\right)$, asymptotically.

3.2.2 Sine Inverse ($\sin^{-1}$) Transformation

Sine-inverse is the transformation for stabilizing the variance of a binomial variate.

If p is the observed proportion of successes in a series of n independent trials with constant probability P of success for each trial then we know that the asymptotic distribution of p is asymptotically normal (as $n \rightarrow \infty$) with mean

$$E(p) = P,$$

and

$$Var(p) = \frac{PQ}{n}, \quad Q = 1 - P.$$

Hence $p \sim N\left(P, \frac{PQ}{n}\right)$, as $n \rightarrow \infty$.

In the usual notations

$$\Psi(p) = \frac{PQ}{n} = \frac{P(1-P)}{n}.$$

Using (3.2), the transforming function is

$$\begin{aligned}
&= c\sqrt{n} \int \frac{dP}{\sqrt{P(1-P)}} \\
&= 2c\sqrt{n} \sin^{-1}(\sqrt{P}) \\
&\quad \left[\because \frac{d}{dP} \sin^{-1}(\sqrt{P}) = \frac{1}{\sqrt{1-P}} \cdot \frac{1}{2\sqrt{P}} \right]
\end{aligned}$$

Choosing the constant c such that

$$2c\sqrt{n} = 1 \quad \Rightarrow \quad c = \frac{1}{2\sqrt{n}},$$

we get

$$g(P) = \sin^{-1}(\sqrt{P}).$$

Hence the transformed statistic is

$$g(p) = \sin^{-1}\sqrt{p}.$$

Using (3.1), $g(p)$ has mean $\sin^{-1}\sqrt{P}$ and

$$\begin{aligned}
Var(\sin^{-1}\sqrt{p}) &= [g'(P)]^2 \cdot \Psi(P) \\
&= \left[\frac{1}{\sqrt{1-P}} \cdot \frac{1}{2\sqrt{P}} \right]^2 \cdot \frac{P(1-P)}{n} \\
&= \frac{1}{4n}.
\end{aligned}$$

Or alternatively $Var(\sin^{-1}\sqrt{p}) = c^2 = \left(\frac{1}{2\sqrt{n}}\right)^2 = \frac{1}{4n}$.

Hence $\sin^{-1}\sqrt{p} \sim N\left(\sin^{-1}\sqrt{P}, \frac{1}{4n}\right)$, asymptotically.

3.2.3 Logarithmic Transformation

Log transformation is the transformation for stabilizing the variance of the distribution of sample variance. If s^2 is the sample variance in a sample of size n from normal population with variance σ^2 , then the sampling distribution of s^2 is asymptotically

normal (as $n \rightarrow \infty$) with mean

$$E(s^2) = \sigma^2,$$

and

$$Var(s^2) = \frac{2\sigma^4}{n}.$$

In the usual notations we have

$$\Psi(\sigma^2) = \frac{2\sigma^4}{n}.$$

Using (3.2), the transformation function is

$$\begin{aligned} g(\sigma^2) &= \int \frac{c\sqrt{n}}{\sqrt{2}\sigma^2} d\sigma^2 \\ &= \frac{c\sqrt{n}}{\sqrt{2}} \log_e \sigma^2. \end{aligned}$$

We select c in such a way that

$$\frac{c\sqrt{n}}{\sqrt{2}} = 1 \quad \Rightarrow \quad c = \frac{\sqrt{2}}{\sqrt{n}},$$

so that

$$g(\sigma^2) = \log_e \sigma^2.$$

Hence the transformation for the statistic s^2 is

$$g(s^2) = \log_e s^2,$$

and using (3.1), the transformed statistic is normally distributed with mean

$$g(\sigma^2) = \log_e \sigma^2,$$

and variance

$$\begin{aligned} Var [g (s^2)] &= [g' (\sigma^2)]^2 \cdot \Psi (\sigma^2) \\ &= \left(\frac{1}{\sigma^2} \right)^2 \cdot \frac{2\sigma^4}{n} \\ &= \frac{2}{n}. \end{aligned}$$

Or alternatively

$$Var [g (s^2)] = c^2 = \left(\frac{\sqrt{2}}{\sqrt{n}} \right)^2 = \frac{2}{n}.$$

Hence $\log_e s^2 \sim N (\log_e \sigma^2, \frac{2}{n})$, for large n .

3.2.4 Fisher's Z -Transformation

This transformation is suggested for stabilizing the variance of the distribution of correlation coefficient. If r is the sample correlation coefficient in sampling from a correlated bivariate normal population with correlation coefficient ρ then the asymptotic distribution of r , as $n \rightarrow \infty$ is normal with mean

$$E (r) = \rho,$$

$$Var(r) = \frac{(1 - \rho^2)^2}{n}.$$

In the usual notations we have

$$\Psi (\rho) = \frac{(1 - \rho^2)^2}{n}.$$

Using (3.2), the transformation function is

$$g (\rho) = \int \frac{c\sqrt{n}}{1 - \rho^2} d\rho$$

$$= c\sqrt{n} \cdot \frac{1}{2} \log_e \left(\frac{1+\rho}{1-\rho} \right).$$

We select c in such a way that

$$c\sqrt{n} = 1 \quad \Rightarrow \quad c = \frac{1}{\sqrt{n}},$$

so that

$$g(\rho) = \frac{1}{2} \cdot \log_e \left(\frac{1+\rho}{1-\rho} \right).$$

Hence the transformation for the statistic is

$$g(r) = \frac{1}{2} \cdot \log_e \left(\frac{1+r}{1-r} \right),$$

which is denoted by Z , and using (3.1), the transformed statistic is normally distributed with mean

$$g(\rho) = \frac{1}{2} \cdot \log_e \left(\frac{1+\rho}{1-\rho} \right),$$

and variance

$$\begin{aligned} Var[g(r)] &= [g'(\rho)]^2 \cdot \Psi(\rho) \\ &= \left(\frac{1}{1-\rho^2} \right)^2 \cdot \frac{(1-\rho^2)^2}{n} \\ &= \frac{1}{n}, \text{ for large } n. \end{aligned}$$

Or alternatively

$$Var[g(r)] = c^2 = \left(\frac{1}{\sqrt{n}} \right)^2 = \frac{1}{n}.$$

Hence $Z = \frac{1}{2} \cdot \log_e \left(\frac{1+r}{1-r} \right) \sim N \left(\frac{1}{2} \cdot \log_e \left(\frac{1+\rho}{1-\rho} \right), \frac{1}{n} \right)$, for large n .

3.3 Applications

The transformation of a statistic T to a function $g(T)$, the distribution of $g(T)$ is approximately normal and its asymptotic variance is independent of the population parameter θ . Hence the use of statistic $g(T)$ gives better results and confidence intervals than the original statistic T .